

## Course Syllabus Part II

### CSD 310 Database Development and Use

#### 9-Week Course

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#### Course Resources

**Course Text:**

MySQL Explained; Andrew Comeau; OStraining; 2017

Sams Teach Yourself SQL in 10 Minutes (5<sup>th</sup> edition); Ben Forta; Sams Publishing; 2020

**Required Resources:**

Computer with internet access and file storage. Links to supplemental materials provided within weekly course descriptions.

**Prerequisite:**

CSD 205 Introduction to Programming with Python

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**Course Schedule**

| Week | Module | Topics Covered                          | Readings                                                                             | Assignments                                                                                                        |
|------|--------|-----------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| 1    | 1      | Relational vs. Non-Relational Databases | Lesson 1 (MySQL Explained); Web articles                                             | <ul style="list-style-type: none"><li>• Discussion 1.1</li><li>• Assignment 1.2</li><li>• Assignment 1.3</li></ul> |
| 2    | 2      | Data Modeling                           | Lesson 3 (MySQL Explained); Web articles                                             | <ul style="list-style-type: none"><li>• Discussion 2.1</li><li>• Assignment 2.2</li><li>• Assignment 2.3</li></ul> |
| 3    | 3      | Normalization                           | Lesson 4 (MySQL Explained); Web articles                                             | <ul style="list-style-type: none"><li>• Discussion 3.1</li><li>• Assignment 3.2</li></ul>                          |
| 4    | 4      | MySQL Installation                      | Lesson 2 (MySQL Explained); Web articles                                             | <ul style="list-style-type: none"><li>• Discussion 4.1</li><li>• Assignment 4.2</li></ul>                          |
|      | 5      | SQL Functions<br>ERD Construction       | Lesson 6 (MySQL Explained); Web articles                                             | <ul style="list-style-type: none"><li>• Discussion 5.1</li><li>• Assignment 5.2</li><li>• Assignment 5.3</li></ul> |
| 5    | 6      | Table Creation<br>Python Access         | Lesson 7 (MySQL Explained); Appendix B & D (Sams); Web articles                      | <ul style="list-style-type: none"><li>• Discussion 6.1</li><li>• Assignment 6.2</li></ul>                          |
| 6    | 7      | SQL Queries Using Python                | Lessons 8, 9 (MySQL Explained); Lessons 10, 18, 19 (Sams); Web articles Web articles | <ul style="list-style-type: none"><li>• Discussion 7.1</li><li>• Assignment 7.2</li></ul>                          |

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|---|----|---------------------------------------------------------------|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
|   | 8  | SQL Insert & Delete Using Python                              | Lesson 9 (MySQL Explained); Lessons 11, 13, 16 (Sams); Web articles Web articles | <ul style="list-style-type: none"> <li>• Discussion 8.1</li> <li>• Assignment 8.2</li> </ul>                              |
| 7 | 9  | Case Study Milestone #1: ERD                                  | All Previous Material                                                            | <ul style="list-style-type: none"> <li>• Assignment 9.1</li> </ul>                                                        |
|   | 10 | Case Study Milestone #2: Development                          | All Previous Material                                                            | <ul style="list-style-type: none"> <li>• Assignment 10.1</li> <li>•</li> </ul>                                            |
| 8 | 11 | Case Study Milestone #3: Reports<br>Milestone #4: Peer Review | All Previous Material                                                            | <ul style="list-style-type: none"> <li>• Assignment 11.1</li> <li>• Assignment 11.2</li> </ul>                            |
| 9 | 12 | Case Study Milestone #5: Revisions                            | N/A                                                                              | <ul style="list-style-type: none"> <li>• Discussion 12.1</li> <li>• Assignment 12.2</li> <li>• Assignment 12.3</li> </ul> |

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### Course Activities [Optional]

N/A

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### Grade Breakdown/Criteria

#### Grade component

10 Discussion Posts

15 Assignments

#### Available points

50 pts. each, total of 500 pts.

50 pts. each, total of 750 pts.

Total points possible 1,250 pts.

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### Late Work

Late work is not accepted unless arrangements are made with the instructor for special circumstances.

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### Participation

Students are expected to login often and contribute to the class on a regular basis, including posting to the discussion board, submitting assignments, and participating in group activities as required. If you have specific participation requirements related to your educational funding or student status, you are expected to monitor your own participation to ensure you are in compliance with those requirements.

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**Expectations for Students**

- Students should expect to spend approximately 15 hours per week to complete the activities and assignments in this course.
  - Students will log in as often as needed to complete their assignments and progress through the course.
  - Students will treat their classmates and the instructor with respect and courtesy.
  - Students are responsible for keeping current with the reading assignments and coming to class prepared to discuss the work assigned.
  - Students are responsible for knowing what assignments are due and when.
  - Students will submit only their own work and will not commit plagiarism or other acts of academic dishonesty.
  - Students will contact the instructor as soon as personal problems arise that may affect the student's ability to complete assignments on time.
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**Expectations for Faculty**

- The instructor will treat all students with respect and courtesy.
  - The instructor will make grading criteria clear and follow the criteria scrupulously in evaluating student work.
  - The instructor will provide feedback about student work within 6 days of due dates (or 24 hours prior to the next due date)—feedback that helps the student learn and improve.
  - The instructor will respond to all student phone and email messages within 48 hours.
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